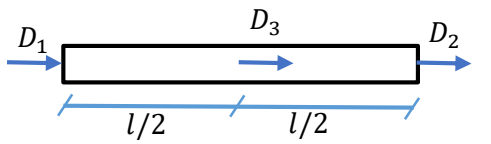
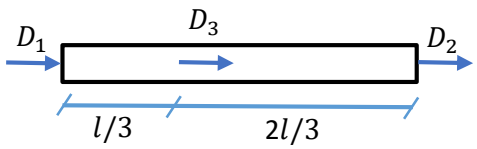
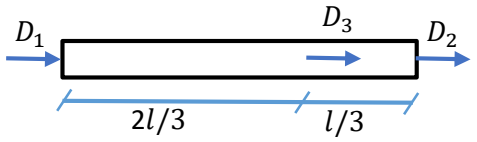
تمرین سری دوم (روش های المان محدود)

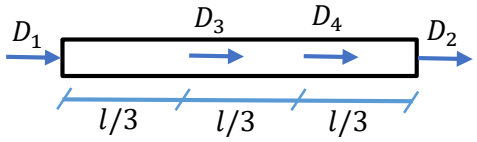
(دانشجو: سبا آزادگان / شماره دانشجویی: ۶۰۲[redacted]۹۷)

1. توابع شکل المان های مرتبه بالای داده شده را با توضیح نوع انتخاب چند جمله ای های تابع میدان به دست آورید.









2. ماتریس سختی دو بخش شده ی المان مرتبه پایین اولر-برنولی را به دست آورید.
3. بخش خمشی ماتریس سختی ۳ المان مرتبه بالا اولر-برنولی را محاسبه کنید.
4. بخش خمشی ماتریس سختی المان مرتبه بالا تیری با دو گره میانی را محاسبه نمایید. (تشویقی)
5. ماتریس سختی برشی المان تیر مرتبه پایین اولر-برنولی محاسبه شود. آیا این ماتریس دارای درایه های یکسان با نتیجه حاصل از قضایای کاستلیانو است؟ (بحث کنید)

حل:

1. توابع شکل المان ها:

با توجه به آن که سه المان اول دارای سه درجه آزادی هستند، تابع میدان هر سه المان شامل ۳ جمله اول مثلث خیام پاسکال میشود. (به دلیل آن که المان ها در جهت x دارای درجه آزادی می باشند، فقط جملات شامل x تا توان ۲ نوشته می شوند.) برای المان چهارم نیز جملات مثلث خیام پاسکال تا توان ۳ انتخاب می شود.

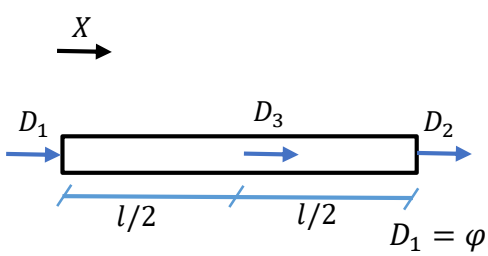
تابع میدان برای سه المان اول دارای سه درجه آزادی:

$$\varphi(x) \overset{RBM}{=} \alpha_1 + \alpha_2 x + \alpha_3 x^2 = [P]\{\alpha\}$$

تابع میدان برای المان چهارم دارای چهار درجه آزادی:

$$\varphi(x) \overset{RBM}{=} \alpha_1 + \alpha_2 x + \alpha_3 x^2 + \alpha_4 x^3 = [P]\{\alpha\}$$

المان اول:



$$D_1 = \varphi_{(x=0)} = \alpha_1 + \alpha_2(0) + \alpha_3(0)^2$$

$$D_2 = \varphi_{(x=l)} = \alpha_1 + \alpha_2(l) + \alpha_3(l)^2$$

$$D_3 = \varphi_{(x=l/2)} = \alpha_1 + \alpha_2(l/2) + \alpha_3(l/2)^2$$

$$[A_E] = \begin{array}{c} \begin{array}{ccc} \alpha_1 & \alpha_2 & \alpha_3 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 \\ 1 & l & l^2 \\ 1 & \frac{l}{2} & \frac{l^2}{8} \end{bmatrix} \end{array} \begin{array}{c} \\ \varphi_1 \\ \varphi_2 \\ \varphi_3 \end{array} \rightarrow [A_E]^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{7}{3l} & -\frac{1}{3l} & \frac{8}{3l} \\ \frac{4}{3l^2} & \frac{4}{3l^2} & -\frac{8}{3l^2} \end{bmatrix}$$

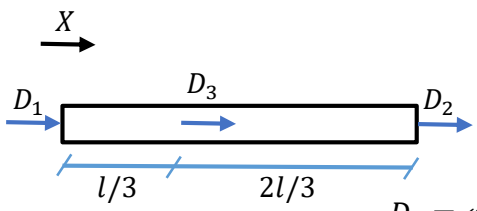
$$[P] = [1 \quad x \quad x^2]$$

$$\boldsymbol{[N_{(x)}] = [P][A_E]^{-1}}$$

$$[N_{1(x)} \quad N_{2(x)} \quad N_{3(x)}] = [1 \quad x \quad x^2] \begin{bmatrix} 1 & 0 & 0 \\ -\frac{7}{3l} & -\frac{1}{3l} & \frac{8}{3l} \\ \frac{4}{3l^2} & \frac{4}{3l^2} & -\frac{8}{3l^2} \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} N_{1(x)} \\ N_{2(x)} \\ N_{3(x)} \end{bmatrix}^T = \begin{array}{ccc} Node1 & Node2 & Node3 \\ \left[1 - \frac{7x}{3l} + \frac{4x^2}{3l^2} \right. & -\frac{x}{3l} + \frac{4x^2}{3l^2} & \left. \frac{8x}{3l} - \frac{8x^2}{3l^2}\right] \end{array}$$

المان دوم:



$$D_1 = \varphi_{(x=0)} = \alpha_1 + \alpha_2(0) + \alpha_3(0)^2$$

$$D_2 = \varphi_{(x=l)} = \alpha_1 + \alpha_2(l) + \alpha_3(l)^2$$

$$D_3 = \varphi_{(x=l/3)} = \alpha_1 + \alpha_2(l/3) + \alpha_3(l/3)^2$$

$$[A_E] = \begin{array}{c} \begin{array}{ccc} \alpha_1 & \alpha_2 & \alpha_3 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 \\ 1 & l & l^2 \\ 1 & \frac{l}{3} & \frac{l^2}{9} \end{bmatrix} \end{array} \begin{array}{c} \\ \varphi_1 \\ \varphi_2 \\ \varphi_3 \end{array} \rightarrow [A_E]^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{4}{l} & -\frac{1}{2l} & \frac{9}{2l} \\ \frac{3}{l^2} & \frac{3}{2l^2} & -\frac{9}{2l^2} \end{bmatrix}$$

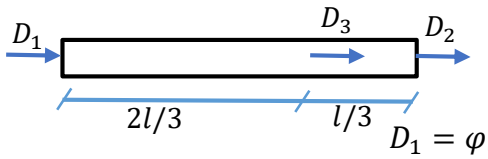
$$[P] = [1 \quad x \quad x^2]$$

$$\boldsymbol{[N_{(x)}] = [P][A_E]^{-1}}$$

$$[N_{1(x)} \quad N_{2(x)} \quad N_{3(x)}] = [1 \quad x \quad x^2] \begin{bmatrix} 1 & 0 & 0 \\ -\frac{4}{l} & -\frac{1}{2l} & \frac{9}{2l} \\ \frac{3}{l^2} & \frac{3}{2l^2} & -\frac{9}{2l^2} \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} N_{1(x)} \\ N_{2(x)} \\ N_{3(x)} \end{bmatrix}^T = \begin{array}{ccc} Node1 & Node2 & Node3 \\ \left[1 - \frac{4x}{l} + \frac{3x^2}{l^2} \right. & -\frac{x}{2l} + \frac{3x^2}{2l^2} & \left. \frac{9x}{2l} - \frac{9x^2}{2l^2}\right] \end{array}$$

المان سوم:



$$D_1 = \varphi_{(x=0)} = \alpha_1 + \alpha_2(0) + \alpha_3(0)^2$$

$$D_2 = \varphi_{(x=l)} = \alpha_1 + \alpha_2(l) + \alpha_3(l)^2$$

$$D_3 = \varphi_{(x=2l/3)} = \alpha_1 + \alpha_2(2l/3) + \alpha_3(2l/3)^2$$

$$[A_E] = \begin{array}{c} \begin{array}{ccc} \alpha_1 & \alpha_2 & \alpha_3 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 \\ 1 & l & l^2 \\ 1 & \frac{2l}{3} & \frac{4l^2}{9} \end{bmatrix} \end{array} \begin{array}{c} \\ \varphi_1 \\ \varphi_2 \\ \varphi_3 \end{array} \rightarrow [A_E]^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{5}{2l} & -\frac{2}{l} & \frac{9}{2l} \\ \frac{3}{2l^2} & \frac{3}{l^2} & -\frac{9}{2l^2} \end{bmatrix}$$

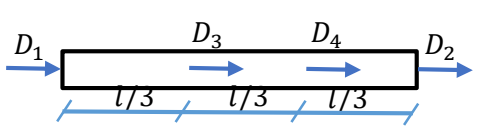
$$[P] = [1 \quad x \quad x^2]$$

$$\boldsymbol{[N_{(x)}] = [P][A_E]^{-1}}$$

$$[N_{1(x)} \quad N_{2(x)} \quad N_{3(x)}] = [1 \quad x \quad x^2] \begin{bmatrix} 1 & 0 & 0 \\ -\frac{5}{2l} & -\frac{2}{l} & \frac{9}{2l} \\ \frac{3}{2l^2} & \frac{3}{l^2} & -\frac{9}{2l^2} \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} N_{1(x)} \\ N_{2(x)} \\ N_{3(x)} \end{bmatrix}^T = \begin{array}{c} \begin{array}{ccc} Node1 & Node2 & Node3 \end{array} \\ \begin{bmatrix} 1 - \frac{5x}{2l} + \frac{3x^2}{2l^2} & -\frac{2x}{l} + \frac{3x^2}{l^2} & \frac{9x}{2l} - \frac{9x^2}{2l^2} \end{bmatrix} \end{array}$$

المان چهارم:



$$D_1 = \varphi_{(x=0)} = \alpha_1 + \alpha_2(0) + \alpha_3(0)^2 + \alpha_4(0)^3$$

$$D_2 = \varphi_{(x=l)} = \alpha_1 + \alpha_2(l) + \alpha_3(l)^2 + \alpha_4(l)^3$$

$$D_3 = \varphi_{(x=l/3)} = \alpha_1 + \alpha_2(l/3) + \alpha_3(l/3)^2 + \alpha_4(l/3)^3$$

$$D_4 = \varphi_{(x=2l/3)} = \alpha_1 + \alpha_2(2l/3) + \alpha_3(2l/3)^2 + \alpha_4(2l/3)^3$$

$$
\begin{array}{cccc} \alpha_1 & \alpha_2 & \alpha_3 & \alpha_4 \end{array}
$$

$$
[A_E] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & l & l^2 & l^3 \\ 1 & \frac{l}{3} & \frac{l^2}{9} & \frac{l^3}{27} \\ 1 & \frac{2l}{3} & \frac{4l^2}{9} & \frac{8l^3}{27} \end{bmatrix} \rightarrow [A_E]^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -\frac{11}{2l} & \frac{1}{l} & \frac{9}{l} & -\frac{9}{2l} \\ \frac{9}{l^2} & -\frac{9}{2l^2} & -\frac{45}{2l^2} & \frac{18}{l^2} \\ -\frac{9}{2l^3} & \frac{9}{2l^3} & \frac{27}{2l^3} & -\frac{27}{2l^3} \end{bmatrix}
$$

$$
[P] = [1 \quad x \quad x^2 \quad x^3]
$$

$$
\boldsymbol{[N_{(x)}] = [P][A_E]^{-1}}
$$

$$
[N_{1(x)} \quad N_{2(x)} \quad N_{3(x)} \quad N_{4(x)}] = [1 \quad x \quad x^2 \quad x^3] \begin{bmatrix} 1 & 0 & 0 & 0 \\ -\frac{11}{2l} & \frac{1}{l} & \frac{9}{l} & -\frac{9}{2l} \\ \frac{9}{l^2} & -\frac{9}{2l^2} & -\frac{45}{2l^2} & \frac{18}{l^2} \\ -\frac{9}{2l^3} & \frac{9}{2l^3} & \frac{27}{2l^3} & -\frac{27}{2l^3} \end{bmatrix}
$$

$$
\rightarrow \begin{bmatrix} N_{1(x)} \\ N_{2(x)} \\ N_{3(x)} \\ N_{4(x)} \end{bmatrix}^T =
$$

$$
\begin{array}{cccc} Node1 & Node2 & Node3 & Node4 \end{array}
$$

$$
\left[1 - \frac{11x}{2l} + \frac{9x^2}{l^2} - \frac{9x^3}{2l^3} \quad \frac{x}{l} - \frac{9x^2}{2l^2} + \frac{9x^3}{2l^3} \quad \frac{9x}{l} - \frac{45x^2}{2l^2} + \frac{27x^3}{2l^3} \quad -\frac{9x}{2l} + \frac{18x^2}{l^2} - \frac{27x^3}{2l^3} \right]
$$

2. ماتریس سختی دو بخش شده ی المان مرتبه پایین اولر-برنولی:

$\varphi(x) = \alpha_1 + \alpha_2 x + \alpha_3 x^2 + \alpha_4 x^3$ $\varphi'_{(x)} = 0 + \alpha_2 + 2\alpha_3 x + 3\alpha_4 x^2$
$D_1 = \varphi_{(x=0)} = \alpha_1$ $D_2 = \varphi'_{(x=0)} = \alpha_2$ $D_3 = \varphi_{(x=l)} = \alpha_1 + \alpha_2 l + \alpha_3 l^2 + \alpha_4 l^3$ $D_4 = \varphi'_{(x=l)} = \alpha_2 + 2\alpha_3 l + 3\alpha_4 l^2$
$[A_E] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & l & l^2 & l^3 \\ 0 & 1 & 2l & 3l^2 \end{bmatrix}$
$[A_E]^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -\frac{3}{l^2} & -\frac{2}{l} & \frac{3}{l^2} & -\frac{1}{l} \\ \frac{2}{l^3} & \frac{1}{l^2} & -\frac{2}{l^3} & \frac{1}{l^2} \end{bmatrix}$
$[P] = [1 \quad x \quad x^2 \quad x^3]$
$[N_{(x)}] = [P][A_E]^{-1}$
$N_{1(x)} = 1 - \frac{3x^2}{l^2} + \frac{2x^3}{l^3}$ $N_{2(x)} = x - \frac{2x^2}{l} + \frac{x^3}{l^2}$ $N_{3(x)} = \frac{3x^2}{l^2} - \frac{2x^3}{l^3}$ $N_{4(x)} = -\frac{x^2}{l} + \frac{x^3}{l^2}$
$N'_{1(x)} = -\frac{2(3x\ )}{l^2} + \frac{3(2x^2)}{l^3}$ $N'_{2(x)} = 1 - \frac{2(2x\ )}{l} + \frac{3(x^2)}{l^2}$ $N'_{3(x)} = \frac{2(3x\ )}{l^2} - \frac{3(2x^2)}{l^3}$ $N'_{4(x)} = -\frac{2(x\ )}{l} + \frac{3(x^2)}{l^2}$

$$N''_{1(x)} = -\frac{2(3)}{l^2} + \frac{6(2x)}{l^3}$$
$$N''_{2(x)} = 1 - \frac{2(2)}{l} + \frac{6(x)}{l^2}$$
$$N''_{3(x)} = \frac{2(3)}{l^2} - \frac{6(2x)}{l^3}$$
$$N''_{4(x)} = -\frac{2(1)}{l} + \frac{6(x)}{l^2}$$

ماتریس سختی برشی

$$[K]_S = GA\begin{bmatrix} \frac{6}{5l} & \frac{1}{10} & -\frac{6}{5l} & \frac{1}{10} \\ \frac{1}{10} & \frac{2l}{15} & -\frac{1}{10} & -\frac{l}{30} \\ -\frac{6}{5l} & -\frac{1}{10} & \frac{6}{5l} & -\frac{1}{10} \\ \frac{1}{10} & -\frac{l}{30} & -\frac{1}{10} & \frac{2l}{15} \end{bmatrix}$$

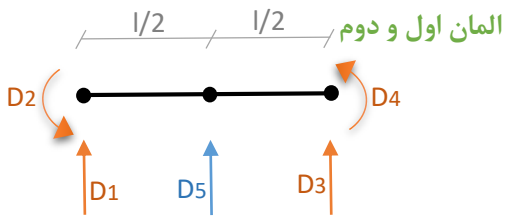
ماتریس سختی خمشی

$$[K]_F = EI\begin{bmatrix} \frac{12(12l^2 + 20l + 15)}{5l^3} & \frac{2(l-3)}{l^2} & -\frac{12}{l^2} & \frac{2(5l+3)}{l^2} \\ \frac{2(l-3)}{l^2} & \frac{4}{l} & -\frac{6}{l^2} & \frac{2}{l} \\ -\frac{12}{l^2} & -\frac{6}{l^2} & \frac{12}{l^3} & -\frac{6}{l^2} \\ \frac{2(5l+3)}{l^2} & \frac{2}{l} & -\frac{6}{l^2} & \frac{4}{l} \end{bmatrix}$$

$$[K]_S = GA\int_0^l \begin{bmatrix} -\frac{6x}{l^2} + \frac{6x^2}{l^3} \\ 1 - \frac{4x}{l} + \frac{3x^2}{l^2} \\ \frac{6x}{l^2} - \frac{6x^2}{l^3} \\ -\frac{2x}{l} + \frac{3x^2}{l^2} \end{bmatrix} \begin{bmatrix} -\frac{6x}{l^2} + \frac{6x^2}{l^3} & 1 - \frac{4x}{l} + \frac{3x^2}{l^2} & \frac{6x}{l^2} - \frac{6x^2}{l^3} & -\frac{2x}{l} + \frac{3x^2}{l^2} \end{bmatrix} dx$$

$$[K]_F = EI\int_0^l \begin{bmatrix} \frac{6}{l^2} + \frac{12x^2}{l^3} \\ -\frac{4}{l} + \frac{6x}{l^2} \\ \frac{6}{l^2} - \frac{12x}{l^3} \\ -\frac{2}{l} + \frac{6x}{l^2} \end{bmatrix} \begin{bmatrix} \frac{6}{l^2} + \frac{12x^2}{l^3} & -\frac{4}{l} + \frac{6x}{l^2} & \frac{6}{l^2} - \frac{12x}{l^3} & -\frac{2}{l} + \frac{6x}{l^2} \end{bmatrix} dx$$

3. بخش خمشی ماتریس سختی ۳ المان مرتبه بالا اولر-برنولی: $[K]_F = EI\int_0^l [N''(x)]^T [N''(x)]dx$



تابع میدان ۳ المان:

1) $\varphi(x) = \alpha_1 + \alpha_2 x + \alpha_3 x^2 + \alpha_4 x^3 + \alpha_5 x^4$
2) $\varphi(x) = \alpha_1 + \alpha_2 x + \alpha_3 x^2 + \alpha_4 x^3 + \alpha_5 x^5$
3) $\varphi(x) = \alpha_1 + \alpha_2 x + \alpha_3 x^2 + \alpha_4 x^3 + \alpha_5 x^4 + \alpha_6 x^5$

المان اول:	المان دوم:
$\varphi(x) = \alpha_1 + \alpha_2 x + \alpha_3 x^2 + \alpha_4 x^3 + \alpha_5 x^4$ $\varphi'_{(x)} = 0 + \alpha_2 + 2\alpha_3 x + 3\alpha_4 x^2 + 4\alpha_5 x^3$	$\varphi(x) = \alpha_1 + \alpha_2 x + \alpha_3 x^2 + \alpha_4 x^3 + \alpha_5 x^5$ $\varphi'_{(x)} = 0 + \alpha_2 + 2\alpha_3 x + 3\alpha_4 x^2 + 5\alpha_5 x^4$
$D_1 = \varphi_{(x=0)} = \alpha_1$ $D_2 = \varphi'_{(x=0)} = \alpha_2$ $D_3 = \varphi_{(x=l)} = \alpha_1 + \alpha_2 l + \alpha_3 l^2 + \alpha_4 l^3 + \alpha_5 l^4$ $D_4 = \varphi'_{(x=l)} = \alpha_2 + 2\alpha_3 l + 3\alpha_4 l^2 + 4\alpha_5 l^3$ $D_5 = \varphi_{\left(x=\frac{l}{2}\right)} = \alpha_1 + \alpha_2(\frac{l}{2}) + \alpha_3(\frac{l}{2})^2 + \alpha_4(\frac{l}{2})^3 + \alpha_5(\frac{l}{2})^4$	$D_1 = \varphi_{(x=0)} = \alpha_1$ $D_2 = \varphi'_{(x=0)} = \alpha_2$ $D_3 = \varphi_{(x=l)} = \alpha_1 + \alpha_2 l + \alpha_3 l^2 + \alpha_4 l^3 + \alpha_5 l^5$ $D_4 = \varphi'_{(x=l)} = \alpha_2 + 2\alpha_3 l + 3\alpha_4 l^2 + 5\alpha_5 l^4$ $D_5 = \varphi_{\left(x=\frac{l}{2}\right)} = \alpha_1 + \alpha_2(\frac{l}{2}) + \alpha_3(\frac{l}{2})^2 + \alpha_4(\frac{l}{2})^3 + \alpha_5(\frac{l}{2})^5$
$[A_E] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & l & l^2 & l^3 & l^4 \\ 0 & 1 & 2l & 3l^2 & 4l^3 \\ 1 & l/2 & l^2/4 & l^3/8 & l^4/16 \end{bmatrix}$	$[A_E] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & l & l^2 & l^3 & l^5 \\ 0 & 1 & 2l & 3l^2 & 5l^4 \\ 1 & l/2 & l^2/4 & l^3/8 & l^5/32 \end{bmatrix}$
$[A_E]^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -11/l^2 & -4/l & -5/l^2 & 1/l & 16/l^2 \\ 18/l^3 & 5/l^2 & 14/l^3 & -3/l^2 & -32/l^3 \\ -8/l^4 & -2/l^3 & -8/l^4 & 2/l^3 & 16/l^4 \end{bmatrix}$	$[A_E]^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -47/5l^2 & -18/5l & -17/5l^2 & 3/5l & 64/5l^2 \\ 58/5l^3 & 17/5l^2 & 38/5l^3 & -7/5l^2 & -96/5l^3 \\ -16/5l^5 & -4/5l^4 & -16/5l^5 & 4/5l^4 & 32/5l^5 \end{bmatrix}$
$[P] = [1 \quad x \quad x^2 \quad x^3 \quad x^4]$	$[P] = [1 \quad x \quad x^2 \quad x^3 \quad x^5]$
$[N_{(x)}] = [P][A_E]^{-1}$	$[N_{(x)}] = [P][A_E]^{-1}$
$N_{1(x)} = 1 - \frac{11x^2}{l^2} + \frac{18x^3}{l^3} - \frac{8x^4}{l^4}$ $N_{2(x)} = x - \frac{4x^2}{l} + \frac{5x^3}{l^2} - \frac{2x^4}{l^3}$ $N_{3(x)} = -\frac{5x^2}{l^2} + \frac{14x^3}{l^3} - \frac{8x^4}{l^4}$ $N_{4(x)} = \frac{x^2}{l} - \frac{3x^3}{l^2} + \frac{2x^4}{l^3}$ $N_{5(x)} = \frac{16x^2}{l^2} - \frac{32x^3}{l^3} + \frac{16x^4}{l^4}$	$N_{1(x)} = 1 - \frac{47x^2}{5l^2} + \frac{58x^3}{5l^3} - \frac{16x^5}{5l^5}$ $N_{2(x)} = x - \frac{18x^2}{5l} + \frac{17x^3}{5l^2} - \frac{4x^5}{5l^4}$ $N_{3(x)} = -\frac{17x^2}{5l^2} + \frac{38x^3}{5l^3} - \frac{16x^5}{5l^5}$ $N_{4(x)} = \frac{3x^2}{5l} - \frac{7x^3}{5l^2} + \frac{4x^5}{5l^4}$ $N_{5(x)} = \frac{64x^2}{5l^2} - \frac{96x^3}{5l^3} + \frac{32x^5}{5l^5}$

$N'_{1(x)} = -\frac{2(11x)}{l^2} + \frac{3(18x^2)}{l^3} - \frac{4(8x^3)}{l^4}$ $N'_{2(x)} = 1 - \frac{2(4x)}{l} + \frac{3(5x^2)}{l^2} - \frac{4(2x^3)}{l^3}$ $N'_{3(x)} = -\frac{2(5x)}{l^2} + \frac{3(14x^2)}{l^3} - \frac{4(8x^3)}{l^4}$ $N'_{4(x)} = \frac{2(x)}{l} - \frac{3(3x^2)}{l^2} + \frac{4(2x^3)}{l^3}$ $N'_{5(x)} = \frac{2(16x)}{l^2} - \frac{3(32x^2)}{l^3} + \frac{4(16x^3)}{l^4}$	$N'_{1(x)} = -\frac{2(47x)}{5l^2} + \frac{3(58x^2)}{5l^3} - \frac{5(16x^4)}{5l^5}$ $N'_{2(x)} = 1 - \frac{2(18x)}{5l} + \frac{3(17x^2)}{5l^2} - \frac{5(4x^4)}{5l^4}$ $N'_{3(x)} = -\frac{2(17x)}{5l^2} + \frac{3(38x^2)}{5l^3} - \frac{5(16x^4)}{5l^5}$ $N'_{4(x)} = \frac{2(3x)}{5l} - \frac{3(7x^2)}{5l^2} + \frac{5(4x^4)}{5l^4}$ $N'_{5(x)} = \frac{2(64x)}{5l^2} - \frac{3(96x^2)}{5l^3} + \frac{5(32x^4)}{5l^5}$
$N''_{1(x)} = -\frac{2(11)}{l^2} + \frac{6(18x)}{l^3} - \frac{12(8x^2)}{l^4}$ $N''_{2(x)} = -\frac{2(4)}{l} + \frac{6(5x)}{l^2} - \frac{12(2x^2)}{l^3}$ $N''_{3(x)} = -\frac{2(5)}{l^2} + \frac{6(14x)}{l^3} - \frac{12(8x^2)}{l^4}$ $N''_{4(x)} = \frac{2(1)}{l} - \frac{6(3x)}{l^2} + \frac{12(2x^2)}{l^3}$ $N''_{5(x)} = \frac{2(16)}{l^2} - \frac{6(32x)}{l^3} + \frac{12(16x^2)}{l^4}$	$N''_{1(x)} = -\frac{2(47)}{5l^2} + \frac{6(58x)}{5l^3} - \frac{20(16x^3)}{5l^5}$ $N''_{2(x)} = -\frac{2(18)}{5l} + \frac{6(17x)}{5l^2} - \frac{20(4x^3)}{5l^4}$ $N''_{3(x)} = -\frac{2(17)}{5l^2} + \frac{6(38x)}{5l^3} - \frac{20(16x^3)}{5l^5}$ $N''_{4(x)} = \frac{2(3)}{5l} - \frac{6(7x)}{5l^2} + \frac{20(4x^3)}{5l^4}$ $N''_{5(x)} = \frac{2(64)}{5l^2} - \frac{6(96x)}{5l^3} + \frac{20(32x^3)}{5l^5}$

$$[K]_F = EI\int_0^l [N''(x)]^T [N''(x)]dx \quad (1$$

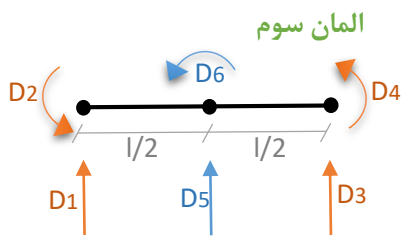
$$[K]_F = EI\int_0^l \begin{bmatrix} -\frac{22}{l^2}+\frac{108x}{l^3}-\frac{96x^2}{l^4} \\ -\frac{8}{l}+\frac{30x}{l^2}-\frac{24x^2}{l^3} \\ -\frac{10}{l^2}+\frac{84x}{l^3}-\frac{96x^2}{l^4} \\ \frac{2}{l}-\frac{18x}{l^2}+\frac{24x^2}{l^3} \\ \frac{32}{l^2}-\frac{192x}{l^3}+\frac{192x^2}{l^4} \end{bmatrix} \begin{bmatrix} -\frac{22}{l^2}+\frac{108x}{l^3}-\frac{96x^2}{l^4} & -\frac{8}{l}+\frac{30x}{l^2}-\frac{24x^2}{l^3} & -\frac{10}{l^2}+\frac{84x}{l^3}-\frac{96x^2}{l^4} & \frac{2}{l}-\frac{18x}{l^2}+\frac{24x^2}{l^3} & \frac{32}{l^2}-\frac{192x}{l^3}+\frac{192x^2}{l^4} \end{bmatrix} dx$$

$$(2$$

$$[K]_F = EI\int_0^l \begin{bmatrix} -\frac{94}{5l^2}+\frac{348x}{5l^3}-\frac{320x^3}{5l^5} \\ -\frac{36}{5l}+\frac{102x}{5l^2}-\frac{80x^3}{5l^4} \\ -\frac{34}{5l^2}+\frac{228x}{5l^3}-\frac{320x^3}{5l^5} \\ \frac{6}{5l}-\frac{42x}{5l^2}+\frac{80x^3}{5l^4} \\ \frac{128}{5l^2}-\frac{576x}{5l^3}+\frac{640x^3}{5l^5} \end{bmatrix} \begin{bmatrix} -\frac{94}{5l^2}+\frac{348x}{5l^3}-\frac{320x^3}{5l^5} & -\frac{36}{5l}+\frac{102x}{5l^2}-\frac{80x^3}{5l^4} & -\frac{34}{5l^2}+\frac{228x}{5l^3}-\frac{320x^3}{5l^5} & \frac{6}{5l}-\frac{42x}{5l^2}+\frac{80x^3}{5l^4} & \frac{128}{5l^2}-\frac{576x}{5l^3}+\frac{640x^3}{5l^5} \end{bmatrix} dx$$

$$[K]_F = EI\begin{bmatrix} \frac{316}{5l^3} & \frac{94}{5l^2} & \frac{196}{5l^3} & -\frac{34}{5l^2} & -\frac{512}{5l^3} \\ \frac{94}{5l^2} & \frac{36}{5l} & \frac{34}{5l^2} & -\frac{6}{5l} & -\frac{128}{5l^2} \\ \frac{196}{5l^3} & \frac{34}{5l^2} & \frac{316}{5l^3} & -\frac{94}{5l^2} & -\frac{512}{5l^3} \\ -\frac{34}{5l^2} & -\frac{6}{5l} & -\frac{94}{5l^2} & \frac{36}{5l} & \frac{128}{5l^2} \\ -\frac{512}{5l^3} & -\frac{128}{5l^2} & -\frac{512}{5l^3} & \frac{128}{5l^2} & \frac{1024}{5l^3} \end{bmatrix}$$

$$[K]_F = EI\begin{bmatrix} \frac{11316}{175l^3} & \frac{3354}{175l^2} & \frac{7116}{175l^3} & -\frac{1254}{175l^2} & -\frac{18432}{175l^3} \\ \frac{3354}{175l^2} & \frac{1276}{175l} & \frac{1254}{175l^2} & -\frac{226}{175l} & -\frac{4608}{175l^2} \\ \frac{7116}{175l^3} & \frac{1254}{175l^2} & \frac{11316}{175l^3} & -\frac{3354}{175l^2} & -\frac{18432}{175l^3} \\ -\frac{1254}{175l^2} & -\frac{226}{175l} & -\frac{3354}{175l^2} & \frac{1276}{175l} & \frac{4608}{175l^2} \\ -\frac{18432}{175l^3} & -\frac{4608}{175l^2} & -\frac{18432}{175l^3} & \frac{4608}{175l^2} & \frac{36864}{175l^3} \end{bmatrix}$$



المان سوم:

$$\varphi(x) = \alpha_1 + \alpha_2 x + \alpha_3 x^2 + \alpha_4 x^3 + \alpha_5 x^4 + \alpha_6 x^5$$

$$\varphi'_{(x)} = 0 + \alpha_2 + 2\alpha_3 x + 3\alpha_4 x^2 + 4\alpha_5 x^3 + 5\alpha_6 x^4$$

$$D_1 = \varphi_{(x=0)} = \alpha_1$$
$$D_2 = \varphi'_{(x=0)} = \alpha_2$$
$$D_3 = \varphi_{(x=l)} = \alpha_1 + \alpha_2 l + \alpha_3 l^2 + \alpha_4 l^3 + \alpha_5 l^4 + \alpha_6 l^5$$
$$D_4 = \varphi'_{(x=l)} = \alpha_2 + 2\alpha_3 l + 3\alpha_4 l^2 + 4\alpha_5 l^3 + 5\alpha_6 l^4$$
$$D_5 = \varphi_{\left(x=\frac{l}{2}\right)} = \alpha_1 + \alpha_2\left(\frac{l}{2}\right) + \alpha_3\left(\frac{l}{2}\right)^2 + \alpha_4\left(\frac{l}{2}\right)^3 + \alpha_5\left(\frac{l}{2}\right)^4 + +\alpha_6\left(\frac{l}{2}\right)^5$$
$$D_6 = \varphi'_{\left(x=\frac{l}{2}\right)} = \alpha_2 + 2\alpha_3\left(\frac{l}{2}\right) + 3\alpha_4\left(\frac{l}{2}\right)^2 + 4\alpha_5\left(\frac{l}{2}\right)^3 + 5\alpha_6\left(\frac{l}{2}\right)^4$$

$$[A_E] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & l & l^2 & l^3 & l^4 & l^5 \\ 0 & 1 & 2l & 3l^2 & 4l^3 & 5l^4 \\ 1 & l/2 & l^2/4 & l^3/8 & l^4/16 & l^5/32 \\ 0 & 1 & l & 3l^2/4 & 4l^3/8 & 5l^4/16 \end{bmatrix} \rightarrow [A_E]^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ -\frac{23}{l^2} & -\frac{6}{l} & \frac{7}{l^2} & -\frac{1}{l} & \frac{16}{l^2} & -\frac{8}{l} \\ \frac{66}{l^3} & \frac{13}{l^2} & -\frac{34}{l^3} & \frac{5}{l^2} & -\frac{32}{l^3} & \frac{32}{l^2} \\ -\frac{68}{l^4} & -\frac{12}{l^3} & \frac{52}{l^4} & -\frac{8}{l^3} & \frac{16}{l^4} & -\frac{40}{l^3} \\ \frac{24}{l^5} & \frac{4}{l^4} & -\frac{24}{l^5} & \frac{4}{l^4} & 0 & \frac{16}{l^4} \end{bmatrix}$$

$$[P] = [1 \quad x \quad x^2 \quad x^3 \quad x^4 \quad x^5]$$

$$[N_{(x)}] = [P][A_E]^{-1}$$

$$N_{1(x)} = 1 - \frac{23x^2}{l^2} + \frac{66x^3}{l^3} - \frac{68x^4}{l^4} + \frac{24x^5}{l^5}$$

$$N_{2(x)} = x - \frac{6x^2}{l} + \frac{13x^3}{l^2} - \frac{12x^4}{l^3} + \frac{4x^5}{l^4}$$

$$N_{3(x)} = \frac{7x^2}{l^2} - \frac{34x^3}{l^3} + \frac{52x^4}{l^4} - \frac{24x^5}{l^5}$$

$$N_{4(x)} = -\frac{x^2}{l} + \frac{5x^3}{l^2} - \frac{8x^4}{l^3} + \frac{4x^5}{l^4}$$

$$N_{5(x)} = \frac{16x^2}{l^2} - \frac{32x^3}{l^3} + \frac{16x^4}{l^4}$$

$$N_{6(x)} = -\frac{8x^2}{l} + \frac{32x^3}{l^2} - \frac{40x^4}{l^3} + \frac{16x^5}{l^4}$$

$$N'_{1(x)} = -\frac{2(23x)}{l^2} + \frac{3(66x^2)}{l^3} - \frac{4(68x^3)}{l^4} + \frac{5(24x^4)}{l^5}$$

$$N'_{2(x)} = 1 - \frac{2(6x)}{l} + \frac{3(13x^2)}{l^2} - \frac{4(12x^3)}{l^3} + \frac{5(4x^4)}{l^4}$$

$$N'_{3(x)} = \frac{2(7x)}{l^2} - \frac{3(34x^2)}{l^3} + \frac{4(52x^3)}{l^4} - \frac{5(24x^4)}{l^5}$$

$$N'_{4(x)} = -\frac{2(x)}{l} + \frac{3(5x^2)}{l^2} - \frac{4(8x^3)}{l^3} + \frac{5(4x^4)}{l^4}$$

$$N'_{5(x)} = \frac{2(16x)}{l^2} - \frac{3(32x^2)}{l^3} + \frac{4(16x^3)}{l^4}$$

$$N'_{6(x)} = -\frac{2(8x)}{l} + \frac{3(32x^2)}{l^2} - \frac{4(40x^3)}{l^3} + \frac{5(16x^4)}{l^4}$$

$$N''_{1(x)} = -\frac{2(23)}{l^2} + \frac{6(66x)}{l^3} - \frac{12(68x^2)}{l^4} + \frac{20(24x^3)}{l^5}$$

$$N''_{2(x)} = -\frac{2(6)}{l} + \frac{6(13x)}{l^2} - \frac{12(12x^2)}{l^3} + \frac{20(4x^3)}{l^4}$$

$$N''_{3(x)} = \frac{2(7)}{l^2} - \frac{6(34x)}{l^3} + \frac{12(52x^2)}{l^4} - \frac{20(24x^3)}{l^5}$$

$$N''_{4(x)} = -\frac{2(1)}{l} + \frac{6(5x)}{l^2} - \frac{12(8x^2)}{l^3} + \frac{20(4x^3)}{l^4}$$

$$N''_{5(x)} = \frac{2(16)}{l^2} - \frac{6(32x)}{l^3} + \frac{12(16x^2)}{l^4}$$

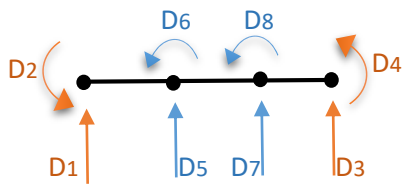
$$N''_{6(x)} = -\frac{2(8)}{l} + \frac{6(32x)}{l^2} - \frac{12(40x^2)}{l^3} + \frac{20(16x^3)}{l^4}$$

$$[K]_F = EI \int_0^l [N''(x)]^T [N''(x)] dx$$

$$[K]_F = EI \int_0^l \begin{bmatrix} -\frac{46}{l^2} + \frac{396x}{l^3} - \frac{816x^2}{l^4} + \frac{480x^3}{l^5} \\ -\frac{12}{l} + \frac{78x}{l^2} - \frac{144x^2}{l^3} + \frac{80x^3}{l^4} \\ \frac{14}{l^2} - \frac{204x}{l^3} + \frac{624x^2}{l^4} - \frac{480x^3}{l^5} \\ -\frac{2}{l} + \frac{30x}{l^2} - \frac{96x^2}{l^3} + \frac{80x^3}{l^4} \\ \frac{32}{l^2} - \frac{192x}{l^3} + \frac{192x^2}{l^4} \\ -\frac{16}{l} + \frac{192x}{l^2} - \frac{480x^2}{l^3} + \frac{320x^3}{l^4} \end{bmatrix} \begin{bmatrix} -\frac{46}{l^2} + \frac{396x}{l^3} - \frac{816x^2}{l^4} + \frac{480x^3}{l^5} & -\frac{12}{l} + \frac{78x}{l^2} - \frac{144x^2}{l^3} + \frac{80x^3}{l^4} & \frac{14}{l^2} - \frac{204x}{l^3} + \frac{624x^2}{l^4} - \frac{480x^3}{l^5} & -\frac{2}{l} + \frac{30x}{l^2} - \frac{96x^2}{l^3} + \frac{80x^3}{l^4} & \frac{32}{l^2} - \frac{192x}{l^3} + \frac{192x^2}{l^4} & -\frac{16}{l} + \frac{192x}{l^2} - \frac{480x^2}{l^3} + \frac{320x^3}{l^4} \end{bmatrix} dx$$

$$[K]_F = EI \begin{bmatrix} \frac{5092}{35l^3} & \frac{1138}{35l^2} & -\frac{1508}{35l^3} & \frac{242}{35l^2} & -\frac{512}{5l^3} & \frac{384}{7l^2} \\ \frac{1138}{35l^2} & \frac{332}{35l} & -\frac{242}{35l^2} & \frac{38}{35l} & -\frac{128}{5l^2} & \frac{64}{7l} \\ -\frac{1508}{35l^3} & -\frac{242}{35l^2} & \frac{5092}{35l^3} & -\frac{1138}{35l^2} & -\frac{512}{5l^3} & -\frac{384}{7l^2} \\ \frac{242}{35l^2} & \frac{38}{35l} & -\frac{1138}{35l^2} & \frac{332}{35l} & \frac{128}{5l^2} & \frac{64}{7l} \\ -\frac{512}{5l^3} & -\frac{128}{5l^2} & -\frac{512}{5l^3} & \frac{128}{5l^2} & \frac{1024}{5l^3} & 0 \\ \frac{384}{7l^2} & \frac{64}{7l} & -\frac{384}{7l^2} & \frac{64}{7l} & 0 & \frac{256}{7l} \end{bmatrix}$$

4. بخش خمشی ماتریس سختی المان مرتبه بالا تیری با دو گره میانی:



$$\boldsymbol{\varphi(x) = \alpha_1 + \alpha_2 x + \alpha_3 x^2 + \alpha_4 x^3 + \alpha_5 x^4 + \alpha_6 x^5 + \alpha_7 x^6 + \alpha_8 x^7}$$

$$\varphi'_{(x)} = 0 + \alpha_2 + 2\alpha_3 x + 3\alpha_4 x^2 + 4\alpha_5 x^3 + 5\alpha_6 x^4 + 6\alpha_7 x^5 + 7\alpha_8 x^6$$

$$D_1 = \varphi_{(x=0)} = \alpha_1$$

$$D_2 = \varphi'_{(x=0)} = \alpha_2$$

$$D_3 = \varphi_{(x=l)} = \alpha_1 + \alpha_2 l + \alpha_3 l^2 + \alpha_4 l^3 + \alpha_5 l^4 + \alpha_6 l^5 + \alpha_7 l^6 + \alpha_8 l^7$$

$$D_4 = \varphi'_{(x=l)} = \alpha_2 + 2\alpha_3 l + 3\alpha_4 l^2 + 4\alpha_5 l^3 + 5\alpha_6 l^4 + 6\alpha_7 l^5 + 7\alpha_8 l^6$$

$$D_5 = \varphi_{\left(x=\frac{l}{3}\right)} = \alpha_1 + \alpha_2\left(\frac{l}{3}\right) + \alpha_3\left(\frac{l}{3}\right)^2 + \alpha_4\left(\frac{l}{3}\right)^3 + \alpha_5\left(\frac{l}{3}\right)^4 + +\alpha_6\left(\frac{l}{3}\right)^5 + \alpha_7\left(\frac{l}{3}\right)^6 + \alpha_8\left(\frac{l}{3}\right)^7$$

$$D_6 = \varphi'_{\left(x=\frac{l}{3}\right)} = \alpha_2 + 2\alpha_3\left(\frac{l}{3}\right) + 3\alpha_4\left(\frac{l}{3}\right)^2 + 4\alpha_5\left(\frac{l}{3}\right)^3 + 5\alpha_6\left(\frac{l}{3}\right)^4 + 6\alpha_7\left(\frac{l}{3}\right)^5 + 7\alpha_8\left(\frac{l}{3}\right)^6$$

$$D_7 = \varphi_{\left(x=\frac{2l}{3}\right)} = \alpha_1 + \alpha_2\left(\frac{2l}{3}\right) + \alpha_3\left(\frac{2l}{3}\right)^2 + \alpha_4\left(\frac{2l}{3}\right)^3 + \alpha_5\left(\frac{2l}{3}\right)^4 + +\alpha_6\left(\frac{2l}{3}\right)^5 + \alpha_7\left(\frac{2l}{3}\right)^6 + \alpha_8\left(\frac{2l}{3}\right)^7$$

$$D_8 = \varphi'_{\left(x=\frac{2l}{3}\right)} = \alpha_2 + 2\alpha_3\left(\frac{2l}{3}\right) + 3\alpha_4\left(\frac{2l}{3}\right)^2 + 4\alpha_5\left(\frac{2l}{3}\right)^3 + 5\alpha_6\left(\frac{2l}{3}\right)^4 + 6\alpha_7\left(\frac{2l}{3}\right)^5 + 7\alpha_8\left(\frac{2l}{3}\right)^6$$

$$[\boldsymbol{A_E}] =$$

1	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0
1	l	l^2	l^3	l^4	l^5	l^6	l^7
0	1	$2l$	$3l^2$	$4l^3$	$5l^4$	$6l^5$	$7l^6$
1	$l/3$	$l^2/9$	$l^3/27$	$l^4/81$	$l^5/243$	$l^6/729$	$l^7/2187$
0	1	$2l/3$	$3l^2/9$	$4l^3/27$	$5l^4/81$	$6l^5/243$	$7l^6/729$
1	$2l/3$	$4l^2/9$	$8l^3/27$	$16l^4/81$	$32l^5/243$	$64l^6/729$	$128l^7/2187$
0	1	$4l/3$	$2l^2/3$	$32l^3/27$	$80l^4/81$	$192l^5/243$	$448l^6/729$

$$[\boldsymbol{A_E}]^{-1} =$$

1	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0
$-\frac{18741}{676l^2}$	$-\frac{1981}{338l}$	$-\frac{114}{169l^2}$	$\frac{11}{169l}$	$\frac{4374}{169l^2}$	$-\frac{675}{169l}$	$\frac{1701}{676l^2}$	$\frac{27}{169l}$
$-\frac{1691}{338l^3}$	$-\frac{193}{338l^2}$	$\frac{238}{169l^3}$	$-\frac{20}{169l^2}$	$-\frac{486}{169l^3}$	$-\frac{432}{169l^2}$	$\frac{2187}{338l^3}$	$-\frac{513}{338l^2}$
$\frac{160011}{338l^4}$	$\frac{11277}{169l^3}$	$\frac{3222}{169l^4}$	$-\frac{1377}{676l^3}$	$-\frac{56619}{169l^4}$	$\frac{86157}{676l^3}$	$-\frac{53217}{338l^4}$	$\frac{918}{169l^3}$
$-\frac{204831}{169l^5}$	$-\frac{27450}{169l^4}$	$-\frac{57663}{676l^5}$	$\frac{6111}{676l^4}$	$\frac{522207}{676l^5}$	$-\frac{243081}{676l^4}$	$\frac{88695}{169l^5}$	$-\frac{1539}{169l^4}$
$\frac{781083}{676l^6}$	$\frac{50787}{338l^5}$	$\frac{39933}{338l^6}$	$-\frac{8667}{676l^5}$	$-\frac{222345}{338l^6}$	$\frac{248103}{676l^5}$	$-\frac{416259}{676l^6}$	$\frac{1215}{169l^5}$
$-\frac{130167}{338l^7}$	$-\frac{16605}{338l^6}$	$-\frac{34911}{676l^7}$	$\frac{3969}{676l^6}$	$\frac{133407}{676l^7}$	$-\frac{86751}{676l^6}$	$\frac{80919}{338l^7}$	$-\frac{729}{338l^6}$

$$[P] = [1 \quad x \quad x^2 \quad x^3 \quad x^4 \quad x^5 \quad x^6 \quad x^7]$$

$$[N_{(x)}] = [P][A_E]^{-1}$$

$$N_{1(x)} = 1 - \frac{\mathbf{18741}x^2}{\mathbf{676}l^2} - \frac{\mathbf{1691}x^3}{\mathbf{338}l^3} + \frac{\mathbf{160011}x^4}{\mathbf{338}l^4} - \frac{\mathbf{204831}x^5}{\mathbf{169}l^5} + \frac{\mathbf{781083}x^6}{\mathbf{676}l^6} - \frac{\mathbf{130167}x^7}{\mathbf{338}l^7}$$

$$N_{2(x)} = x - \frac{1981x^2}{338l} - \frac{193x^3}{338l^2} + \frac{11277x^4}{169l^3} - \frac{27450x^5}{169l^4} + \frac{50787x^6}{338l^5} - \frac{16605x^7}{338l^6}$$

$$N_{3(x)} = -\frac{114x^2}{169l^2} + \frac{238x^3}{169l^3} + \frac{3222x^4}{169l^4} - \frac{57663x^5}{676l^5} + \frac{39933x^6}{338l^6} - \frac{34911x^7}{676l^7}$$

$$N_{4(x)} = \frac{11x^2}{169l} - \frac{20x^3}{169l^2} - \frac{1377x^4}{676l^3} + \frac{6111x^5}{676l^4} - \frac{8667x^6}{676l^5} + \frac{3969x^7}{676l^6}$$

$$N_{5(x)} = \frac{4374x^2}{169l^2} - \frac{486x^3}{169l^3} - \frac{56619x^4}{169l^4} + \frac{522207x^5}{676l^5} - \frac{222345x^6}{338l^6} + \frac{133407x^7}{676l^7}$$

$$N_{6(x)} = -\frac{675x^2}{169l} - \frac{432x^3}{169l^2} + \frac{86157x^4}{676l^3} - \frac{243081x^5}{676l^4} + \frac{248103x^6}{676l^5} - \frac{86751x^7}{676l^6}$$

$$N_{7(x)} = \frac{1701x^2}{676l^2} + \frac{2187x^3}{338l^3} - \frac{53217x^4}{338l^4} + \frac{88695x^5}{169l^5} - \frac{416259x^6}{676l^6} + \frac{80919x^7}{338l^7}$$

$$N_{8(x)} = \frac{27x^2}{169l} - \frac{513x^3}{338l^2} + \frac{918x^4}{169l^3} - \frac{1539x^5}{169l^4} + \frac{1215x^6}{169l^5} - \frac{729x^7}{338l^6}$$

$$N''_{1(x)} = -\frac{2(18741)}{676l^2} - \frac{6(1691x)}{338l^3} + \frac{12(160011x^2)}{338l^4} - \frac{20(204831x^3)}{169l^5} + \frac{30(781083x^4)}{676l^6} - \frac{42(130167x^5)}{338l^7}$$

$$N''_{2(x)} = -\frac{2(1981)}{338l} - \frac{6(193x)}{338l^2} + \frac{12(11277x^2)}{169l^3} - \frac{20(27450x^3)}{169l^4} + \frac{30(50787x^4)}{338l^5} - \frac{42(16605x^5)}{338l^6}$$

$$N''_{3(x)} = -\frac{2(114)}{169l^2} + \frac{6(238x)}{169l^3} + \frac{12(3222x^2)}{169l^4} - \frac{20(57663x^3)}{676l^5} + \frac{30(39933x^4)}{338l^6} - \frac{42(34911x^5)}{676l^7}$$

$$N''_{4(x)} = \frac{2(11)}{169l} - \frac{6(20x)}{169l^2} - \frac{12(1377x^2)}{676l^3} + \frac{20(6111x^3)}{676l^4} - \frac{30(8667x^4)}{676l^5} + \frac{42(3969x^5)}{676l^6}$$

$$N''_{5(x)} = \frac{2(4374)}{169l^2} - \frac{6(486x)}{169l^3} - \frac{12(56619x^2)}{169l^4} + \frac{20(522207x^3)}{676l^5} - \frac{30(222345x^4)}{338l^6} + \frac{42(133407x^5)}{676l^7}$$

$$N''_{6(x)} = -\frac{2(675)}{169l} - \frac{6(432x)}{169l^2} + \frac{12(86157x^2)}{676l^3} - \frac{20(243081x^3)}{676l^4} + \frac{30(248103x^4)}{676l^5} - \frac{42(86751x^5)}{676l^6}$$

$$N''_{7(x)} = \frac{2(1701)}{676l^2} + \frac{6(2187x)}{338l^3} - \frac{12(53217x^2)}{338l^4} + \frac{20(88695x^3)}{169l^5} - \frac{30(416259x^4)}{676l^6} + \frac{42(80919x^5)}{338l^7}$$

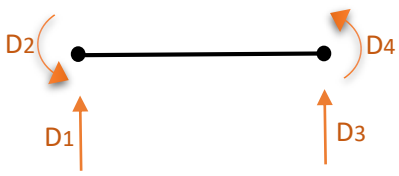
$$N''_{8(x)} = \frac{2(27)}{169l} - \frac{6(513x)}{338l^2} + \frac{12(918x^2)}{169l^3} - \frac{20(1539x^3)}{169l^4} + \frac{30(1215x^4)}{169l^5} - \frac{42(729x^5)}{338l^6}$$

$$[K]_F = EI \int_0^l [N''(x)]^T [N''(x)] dx$$

$$[K]_F = EI$$

$\frac{13851551757}{10995985l^3}$	$\frac{6996521097}{43983940l^2}$						
$\frac{6996521097}{43983940l^2}$	$\frac{341020202}{10995985l}$			sym			
$\frac{61062013293}{175935760l^3}$	$\frac{445518279}{21991970l^2}$	$\frac{3698655762}{10995985l^3}$					
$-\frac{8126805207}{175935760l^2}$	$\frac{191071141}{87967880l}$	$-\frac{642448293}{10995985l^2}$	$\frac{141928492}{10995985l}$				
$-\frac{83354231061}{175935760l^3}$	$-\frac{744907399}{10995985l^2}$	$\frac{3185490159}{43983940l^3}$	$-\frac{308719121}{43983940l^2}$	$\frac{12333508661}{21991970l^3}$			
$\frac{78371339157}{175935760l^2}$	$\frac{626578497}{12566840l}$	$\frac{7034631057}{43983940l^2}$	$-\frac{461365389}{21991970l}$	$-\frac{1622974887}{21991970l^2}$	$\frac{2054889162}{10995985l}$		
$-\frac{3559510899}{3141710l^3}$	$-\frac{4908248379}{43983940l^2}$	$-\frac{132982466121}{175935760l^3}$	$\frac{19638772299}{175935760l^2}$	$-\frac{28055774223}{175935760l^3}$	$-\frac{13360866327}{25133680l^2}$	$\frac{22523178168}{10995985l^3}$	
$\frac{37401102}{10995985l^2}$	$\frac{1204713}{3141710l}$	$\frac{57821607}{43983940l^2}$	$-\frac{2218833}{12566840l}$	$-\frac{3063987}{6283420l^2}$	$\frac{125358111}{87967880l}$	$-\frac{92989053}{21991970l^2}$	$\frac{180792}{10995985l}$

5. ماتریس سختی برشی المان تیر مرتبه پایین اولر-برنولی: $[K]_S = GA\int_0^l [N'(x)]^T[N'(x)]dx$



$$\boldsymbol{\varphi(x) = \alpha_1 + \alpha_2 x + \alpha_3 x^2 + \alpha_4 x^3}$$

$$\varphi'_{(x)} = 0 + \alpha_2 + 2\alpha_3 x + 3\alpha_4 x^2$$

$$D_1 = \varphi_{(x=0)} = \alpha_1$$
$$D_2 = \varphi'_{(x=0)} = \alpha_2$$
$$D_3 = \varphi_{(x=l)} = \alpha_1 + \alpha_2 l + \alpha_3 l^2 + \alpha_4 l^3$$
$$D_4 = \varphi'_{(x=l)} = \alpha_2 + 2\alpha_3 l + 3\alpha_4 l^2$$

$$[A_E] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & l & l^2 & l^3 \\ 0 & 1 & 2l & 3l^2 \end{bmatrix}$$

$$[A_E]^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -\frac{3}{l^2} & -\frac{2}{l} & \frac{3}{l^2} & -\frac{1}{l} \\ \frac{2}{l^3} & \frac{1}{l^2} & -\frac{2}{l^3} & \frac{1}{l^2} \end{bmatrix}$$

$$[P] = [1 \quad x \quad x^2 \quad x^3]$$

$$[N_{(x)}] = [P][A_E]^{-1}$$

$N_{1(x)} = 1 - \frac{3x^2}{l^2} + \frac{2x^3}{l^3}$	$N'_{1(x)} = -\frac{2(3x\ )}{l^2} + \frac{3(2x^2)}{l^3}$
$N_{2(x)} = x - \frac{2x^2}{l} + \frac{x^3}{l^2}$	$N'_{2(x)} = 1 - \frac{2(2x\ )}{l} + \frac{3(x^2)}{l^2}$
$N_{3(x)} = \frac{3x^2}{l^2} - \frac{2x^3}{l^3}$	$N'_{3(x)} = \frac{2(3x\ )}{l^2} - \frac{3(2x^2)}{l^3}$
$N_{4(x)} = -\frac{x^2}{l} + \frac{x^3}{l^2}$	$N'_{4(x)} = -\frac{2(x\ )}{l} + \frac{3(x^2)}{l^2}$

$$\boldsymbol{[K]_S = GA\int_0^l [N'(x)]^T[N'(x)]dx}$$

$$[K]_S = GA\int_0^l \begin{bmatrix} -\frac{6x}{l^2} + \frac{6x^2}{l^3} \\ 1 - \frac{4x}{l} + \frac{3x^2}{l^2} \\ \frac{6x}{l^2} - \frac{6x^2}{l^3} \\ -\frac{2x}{l} + \frac{3x^2}{l^2} \end{bmatrix} \begin{bmatrix} -\frac{6x}{l^2} + \frac{6x^2}{l^3} & 1 - \frac{4x}{l} + \frac{3x^2}{l^2} & \frac{6x}{l^2} - \frac{6x^2}{l^3} & -\frac{2x}{l} + \frac{3x^2}{l^2} \end{bmatrix} dx$$

$$[K]_S = GA\begin{bmatrix} \frac{6}{5l} & \frac{1}{10} & -\frac{6}{5l} & \frac{1}{10} \\ \frac{1}{10} & \frac{2l}{15} & -\frac{1}{10} & -\frac{l}{30} \\ -\frac{6}{5l} & -\frac{1}{10} & \frac{6}{5l} & -\frac{1}{10} \\ \frac{1}{10} & -\frac{l}{30} & -\frac{1}{10} & \frac{2l}{15} \end{bmatrix}$$

تفاوت این ماتریس با ماتریس حاصل از قضایای کاستلیانو:

رابطه ی حاکم در قضایای کاستلیانو برای بدست آوردن ماتریس سختی، تاثیر همزمان خمش و برش است در صورتی که در ماتریس سختی تیر مرتبه پایین اولر-برنولی تنها اثر برش لحاظ شده است، در واقع ماتریس سختی به دو بخش سختی برشی و سختی خمشی تقسیم شده است.

در قضایای کاستلیانو با اعمال تغییر جزئی و محاسبه معادله لنگر و برش و مشتق گیری سختی محاسبه می شود ولی در تیر اویلر-برنولی با تعریف یک تابع شکل است که دقتش به تعداد گره های انتخاب شده و چندجمله ای هایی که از مثلث خیام پاسکال حاصل می شود بستگی دارد.